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Operating systems

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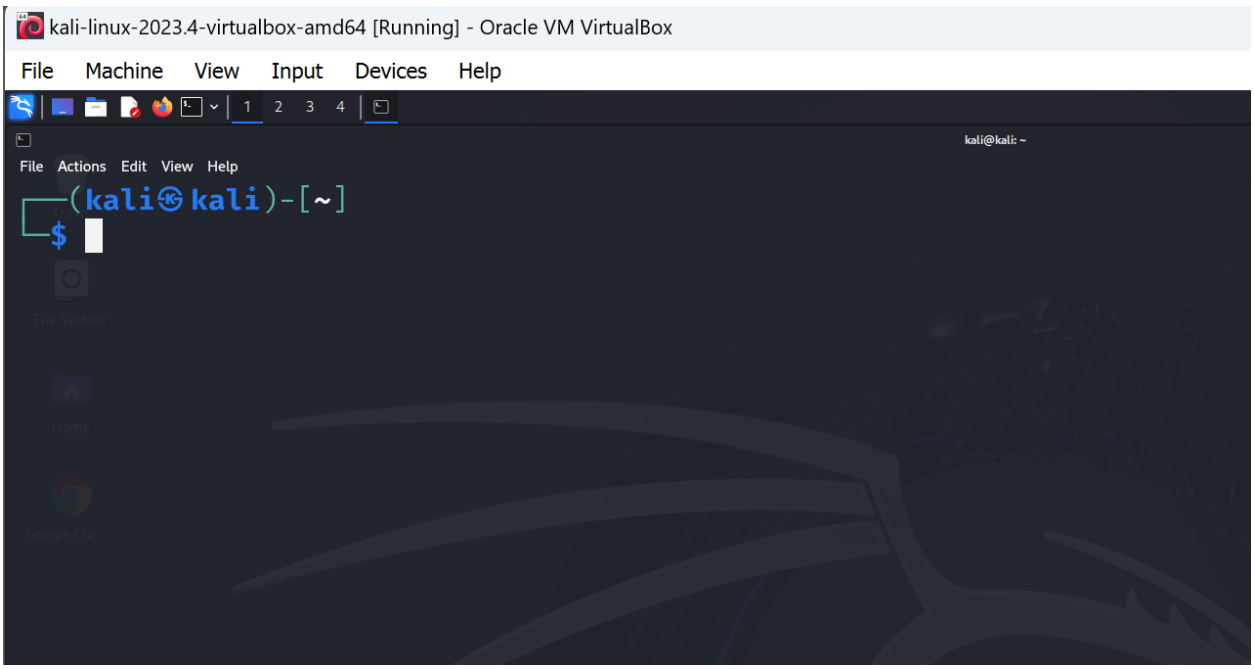
Lab 03- Swapping in Linux

Introduction:

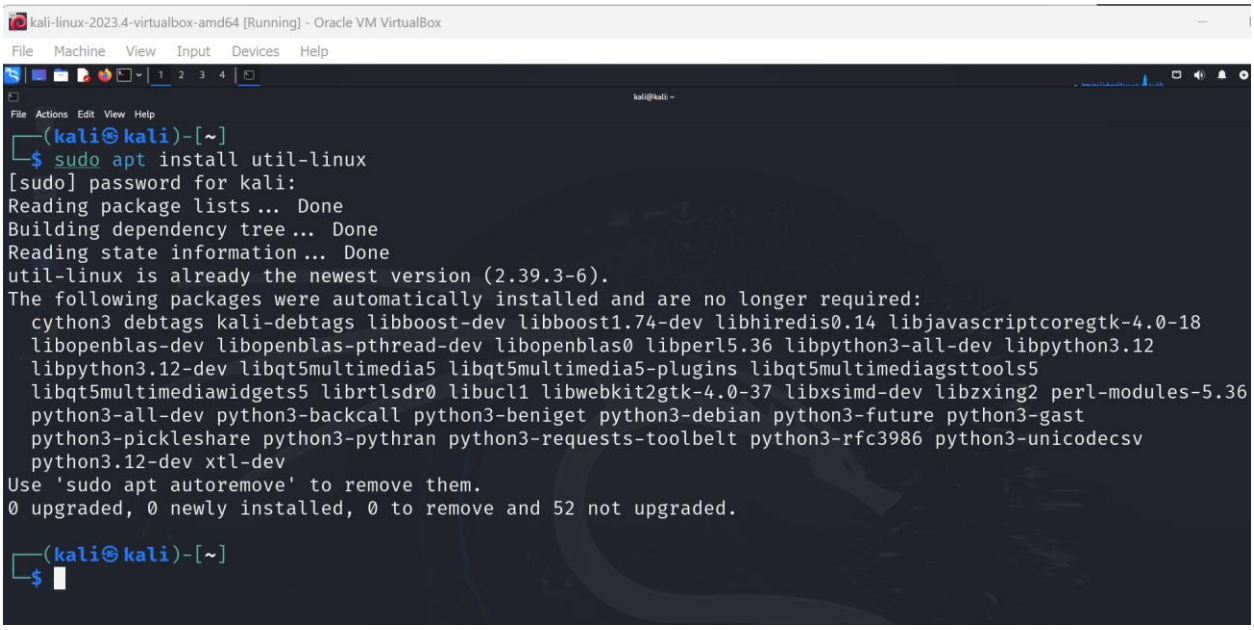
In this lab, I will be using hands on techniques and mastering swapping in the Linux system. This will help in knowing how to use Linux tools and commands with a cybersecurity emphasis. I will be using my Linux on my vm and the slides presentent by the professor.

Objective: To manage swapping in Linux.

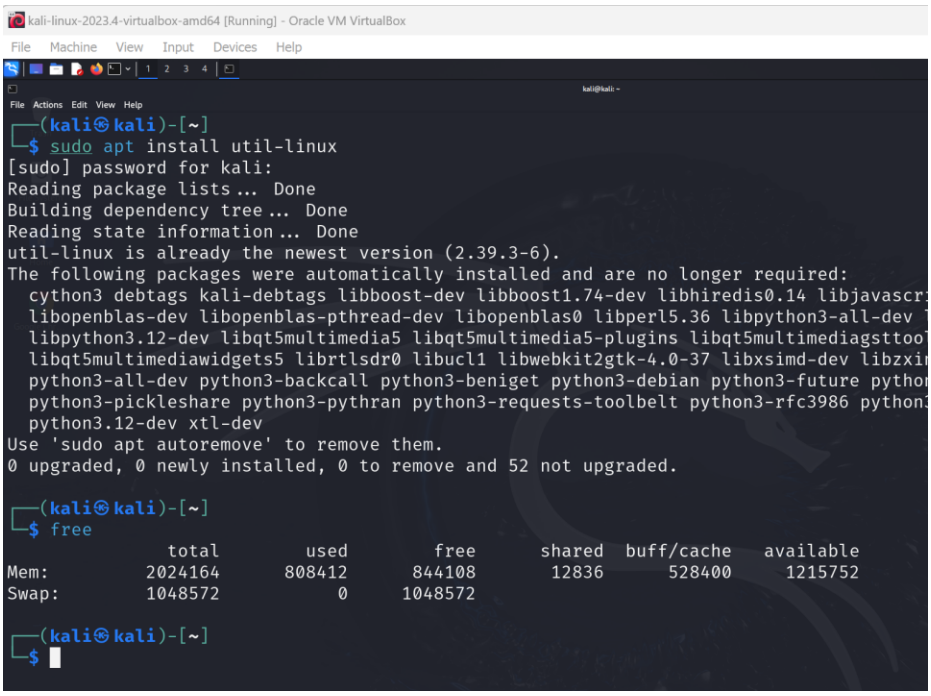
Step 1: In this step, I used the instructed commands to open a Linux terminal. I used Ctrl+Alt+T and then I zoomed in by using Ctrl++.



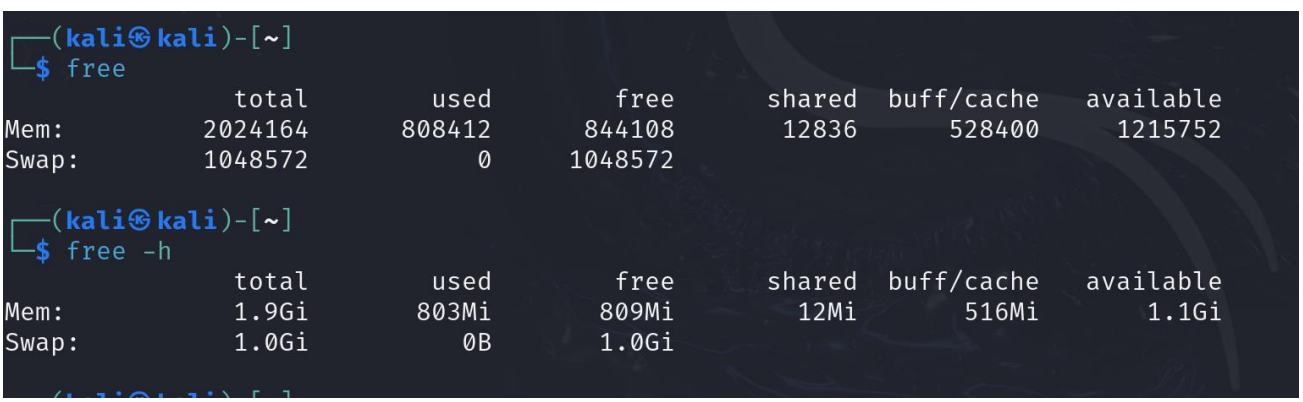
Step 2: In this step, I successfully used the following command to install the tool on my Linux system.



Step 3: In this step, I used the >>free<< command to help me see the current the memory info on my system.



Step 4: I used the >> free –h<< command to have another view of the memory info in my system.



Step 5: After using the >>sudo swapon –h<< command, the swap space info oon my system became visible.

```
┌──(kali㉿kali)-[~]
└─$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.9Gi       803Mi       809Mi        12Mi       516Mi       1.1Gi
Swap:         1.0Gi          0B       1.0Gi

┌──(kali㉿kali)-[~]
└─$ sudo swapon -show
Filename                                Type            Size            Used            Priority
/swapfile                               file            1048572         0               -2

┌──(kali㉿kali)-[~]
└─$
```

Step 6: I used the >>cat /proc/swaps<< command, which basically did the same as the previous command.

```
┌──(kali㉿kali)-[~]
└─$ sudo swapon -show
Filename                                Type            Size            Used            Priority
/swapfile                               file            1048572         0               -2

┌──(kali㉿kali)-[~]
└─$ cat /proc/swaps
Filename                                Type            Size            Used            Priority
/swapfile                               file            1048572         0               -2
```

Step 7: On this step, after using the command >>sudo fallocate –l 2G /swapfile<<

It returned that the text file was busy.

```
┌──(kali㉿kali)-[~]
└─$ sudo fallocate -l 2G /swapfile

fallocate: fallocate failed: Text file busy
```

Step 8: In this step, I swap file commands and afterwards made a mistake but executed the right command and listed all the files.

```
┌──(kali㉿kali)-[~/chernandez]
└─$ sudo chmod 600 /swapfile

┌──(kali㉿kali)-[~/chernandez]
└─$ 1s -la
1s: command not found

┌──(kali㉿kali)-[~/chernandez]
└─$ ls -la
total 8
drwxr-xr-x  2 kali kali 4096 Mar 11 20:17 .
drwx------ 22 kali kali 4096 Mar 29 12:47 ..

┌──(kali㉿kali)-[~/chernandez]
└─$
```

Step 9: I set it up and was able to activate it.

```
┌──(kali㉿kali)-[~/chernandez]
└─$ sudo mkswap /swapfile
mkswap: /swapfile: warning: wiping old swap signature.
Setting up swapspace version 1, size = 1024 MiB (1073737728 bytes)
no label, UUID=2e22f811-d165-4ae1-88cb-6c851052d915

┌──(kali㉿kali)-[~/chernandez]
└─$ sudo swapon /swapfile
```

Step 10: I'm able to see the memory. This command helped me in seeing the expanded memory.

```
┌──(kali㉿kali)-[~/chernandez]
└─$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.9Gi       835Mi       623Mi        12Mi       701Mi       1.1Gi
Swap:         1.0Gi          0B       1.0Gi

┌──(kali㉿kali)-[~/chernandez]
└─$ cat /proc/swaps
Filename                                Type            Size            Used            Priority
/swapfile                               file            1048572         0               -2
```

Step 11: The instructed command lets me to change permanently the file. I changed none to swap.

```
kali-linux-2023.4-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File  Machine  View  Input  Devices  Help
1 2 3 4
kali@kali: ~/chernandez
File  Actions  Edit  View  Help
  GNU nano 7.2                                        /etc/fstab
UUID=24dfb9c0-e01e-4ee9-864b-c62095d6eae7       /        ext4    defaults,errors=remour
/swapfile swap swap defaults 0 0
```

Step 12: The swap file was already in the OS.

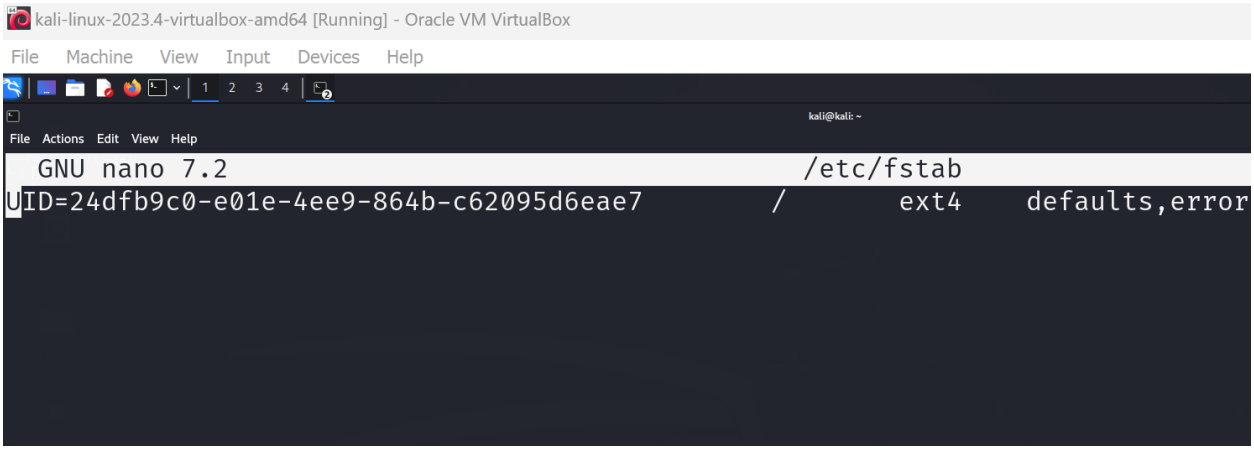
```
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       858Mi       813Mi        22Mi       466Mi       1.1Gi
Swap:          1.0Gi          0B       1.0Gi

┌──(kali㉿kali)-[~]
└─$ cat /proc/swaps
Filename                                Type            Size            Used            Priority
/swapfile                               file            1048572         0               -2
```

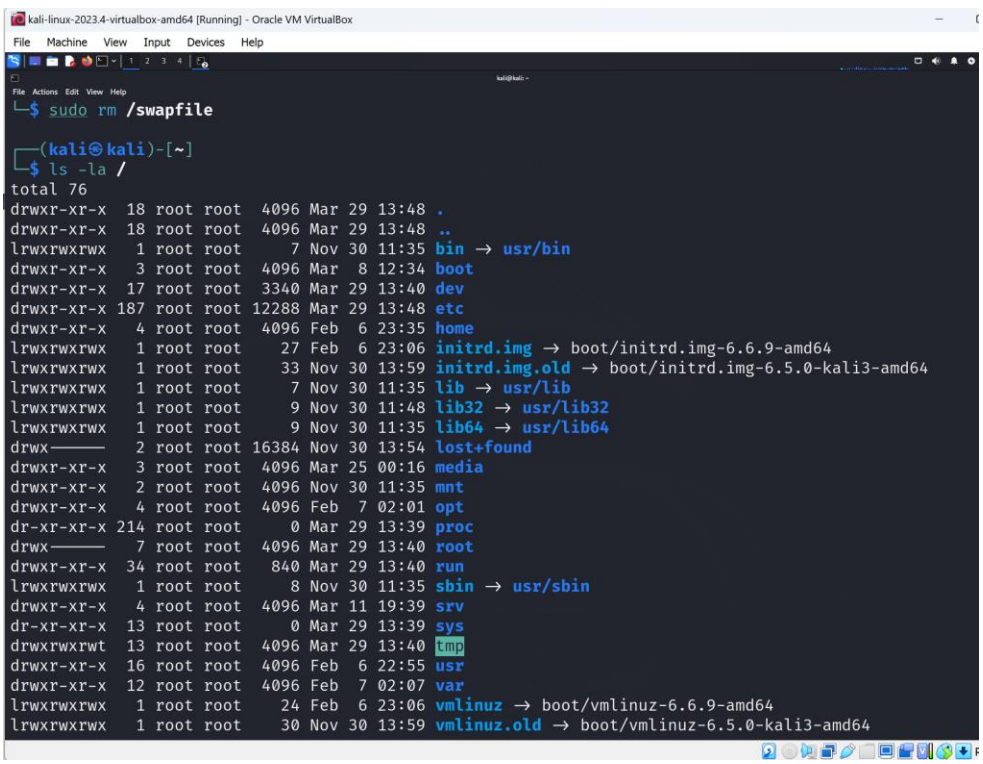
Step 13: The swap file was deactivated.

```
┌──(kali㉿kali)-[~]
└─$ sudo swapoff -v /swapfile
[sudo] password for kali:
swapoff /swapfile
```

Step 14: I deleted the line.



Step 15: After deleting it, I used the >>ls –la /<< to verify is not there anymore.



Questions:

1) What is swapping?

Swapping is a memory management technique used by operating systems to move pages of data between RAM and disk storage. When a system runs out of physical memory (RAM) to hold all the data it needs, it uses swapping to temporarily store data that is not actively being used in RAM onto the disk (swap space). This frees up RAM for other processes, allowing the system to continue running.

2)What is a swap file?

A swap file is used by the operating system as virtual memory when physical memory (RAM) is full. It serves as a temporary storage space for data that is swapped in and out of RAM as needed. Swap files are typically used in systems where the amount of physical memory is limited or when the system needs to handle more processes than the available physical memory can accommodate.

3)How is the swap file priority used?

The swap file priority is used to determine the order in which swap files are used by the operating system when swapping data between RAM and disk. The priority is a value assigned to each swap file, with lower values indicating higher priority. When the system needs to swap data, it first tries to use the swap file with the highest priority that has enough space available. If none of the swap files have enough space, the system may resort to swapping out data from RAM that is not actively being used.

Conclusion:

In this lab, I used multiple commands to better manage swapping in Linux. I have now a better grasp of swapping in the Linux operating system. I practiced verifying the deletion and making a swap permanent. I did make some mistakes but quickly fixed and used the given slides by the professor to orient myself.

References:

Swap. Swap - ArchWiki. (n.d.). https://wiki.archlinux.org/title/swap